

Arab Open University – Bahrain

Faculty of Computer Studies

Smart Inventory Tracking

Inventory Management for Satellite Store at AlBa

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Smart Inventory Tracking

Satellite Store

AlBa

~ Introduction ~

- ❖ Maintenance stores manage large numbers of tools.
- ❖ Many stores still use Excel or handwritten inventory records
- ❖ Manual tracking leads to errors and inefficiency
 - ❖ No real-time visibility of inventory levels
 - ❖ Poor visibility causes stock shortages and overstocking
 - ❖ Time consuming and difficult to manage large inventories
 - ❖ Missing data and duplicated entries
 - ❖ Can cause delays in maintenance operations

~ Project Aims ~

- ❖ Develop a smart, digital system to replace manual inventory tracking for a single store in AlBa
- ❖ Monitor material movement and stock levels
- ❖ Improve accuracy, efficiency and visibility of store inventory
- ❖ Support better planning and cost reduction

~ Research Findings ~

- ❖ Many small stores still rely on spreadsheets – lack automation, alerts and dashboards
- ❖ Many smart inventory tracking systems share core features
 - ❖ Real-time Updates
 - ❖ Barcode Scanning
 - ❖ Low-Stock Alerts
 - ❖ Dashboards
 - ❖ Centralised Logs

~ Research Findings ~

- ❖ Existing systems focus on large-scale or retail environments
- ❖ ERP systems (e.g., Odoo) are complex and costly for single stores
 - ❖ Fully integrated with accounting, sales and manufacturing modules.
 - ❖ Includes complex stock valuation methods and advanced warehouse routing.
 - ❖ Large-scale architecture, advanced workflow management, higher technical setup requirements.
- ❖ Need for a simple, low-cost system for single maintenance stores

~ Suggested Solution ~

- ❖ Smart Inventory Tracking System with barcode scanning
- ❖ Central SQL database for accurate transaction logging
- ❖ Real-time stock updates for withdrawals and restocking
- ❖ Automated alerts for low or overstock situations
- ❖ Dashboard with usage trends
- ❖ Simple, user-friendly interface with minimal IT dependency

~ Target Users ~



Storekeepers & Maintenance Staff

- ~ Faster and easier stock recording
- ~ Reduced manual work and errors



Inventory Managers

- ~ Clear stock visibility and usage trends
- ~ Better planning and decision-making



Company

- ~ Reduced material waste and shortages
- ~ Improved operational efficiency

~ Requirement Elicitation ~



Meeting & emails with
AlBa supervisor



Observation of current
Excel-based system



Review of similar
inventory systems

~ Functional Requirements ~



Central SQL
database for all
records



Barcode-based stock
withdrawal and
restocking



Real-time update of
inventory levels



Transaction history
and usage trends



Low-stock and
overstock alerts



Add, edit and
manage item
categories



Search by item name
or category



Track items sent for
repair



Allow stock
reservation for
planned
maintenance jobs

~ Non Functional Requirements ~



Usability: Easy to use
for non-technical staff



Performance: Fast,
real-time data updates



Maintainability: Simple
to update and fix



Scalability: Grows with
more stores and
inventory



Reliability: Accurate
stock tracking



Security: Safe and
secure data storage



Auditability: All
actions fully traceable



Offline Capability:
Scan offline, sync later

~ Software Development Model ~

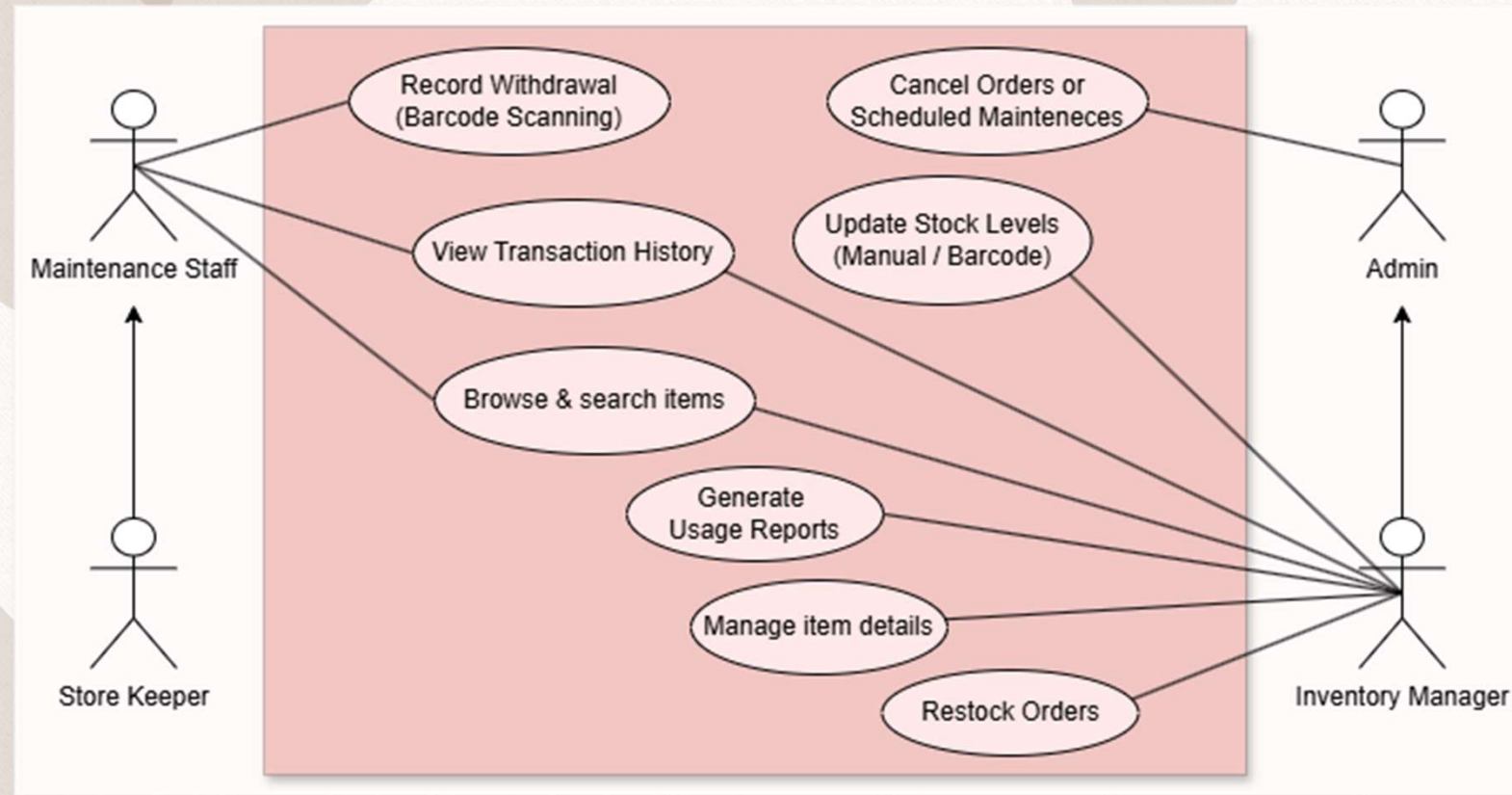
Agile development methodology – Advantages

- ❖ Incremental Delivery – System developed in small iterative modules
- ❖ Feedback Based Development – Continuous feedback from store staff
- ❖ Flexibility – Allows changes during development (requirements evolve)
- ❖ Early detection of issues – Continuous testing
- ❖ Risk reduction – Reduces risk of errors and improves usability

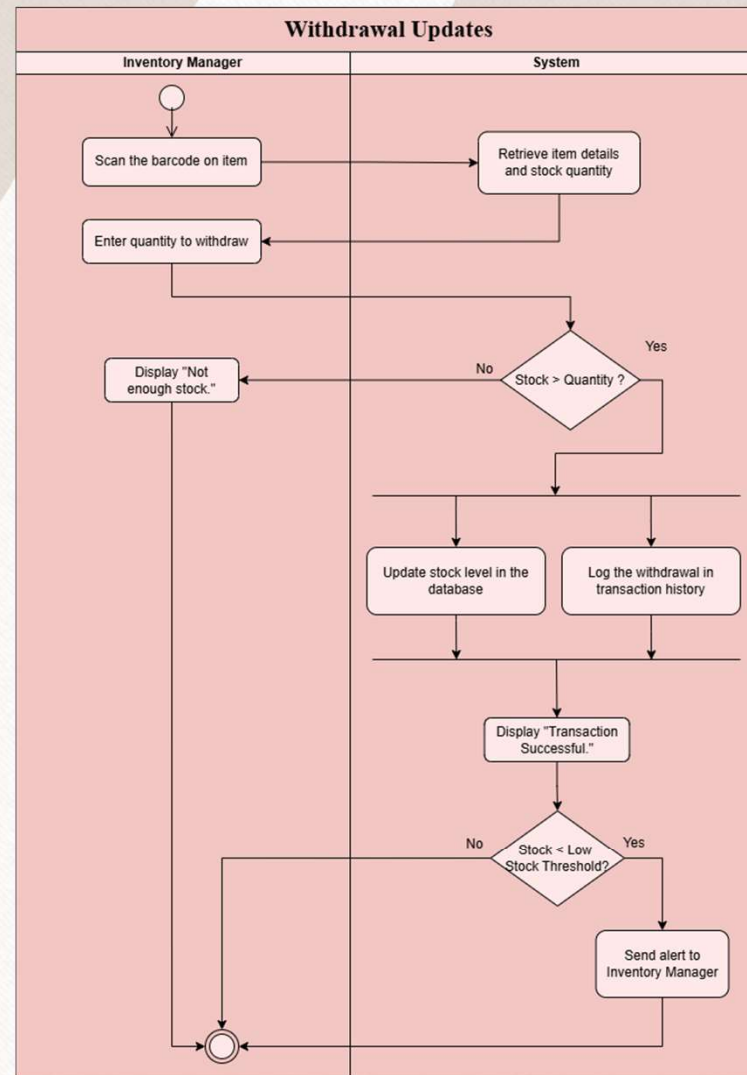


~ Analysis & Design Diagrams ~

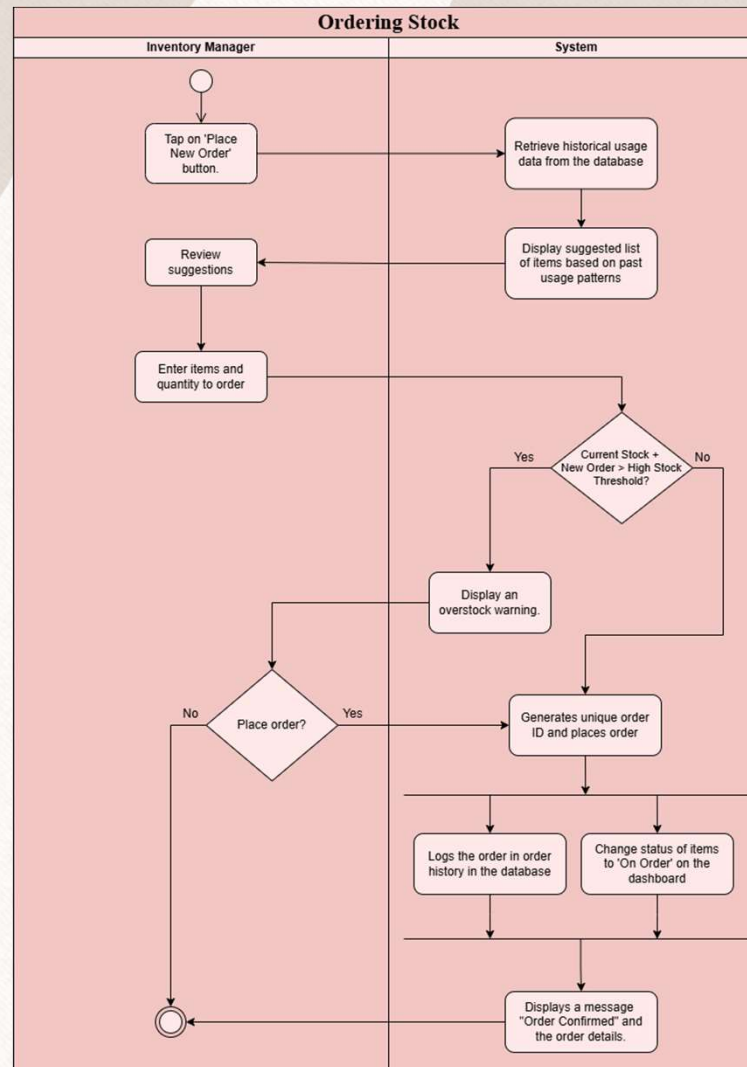
~ Use Case Diagram ~



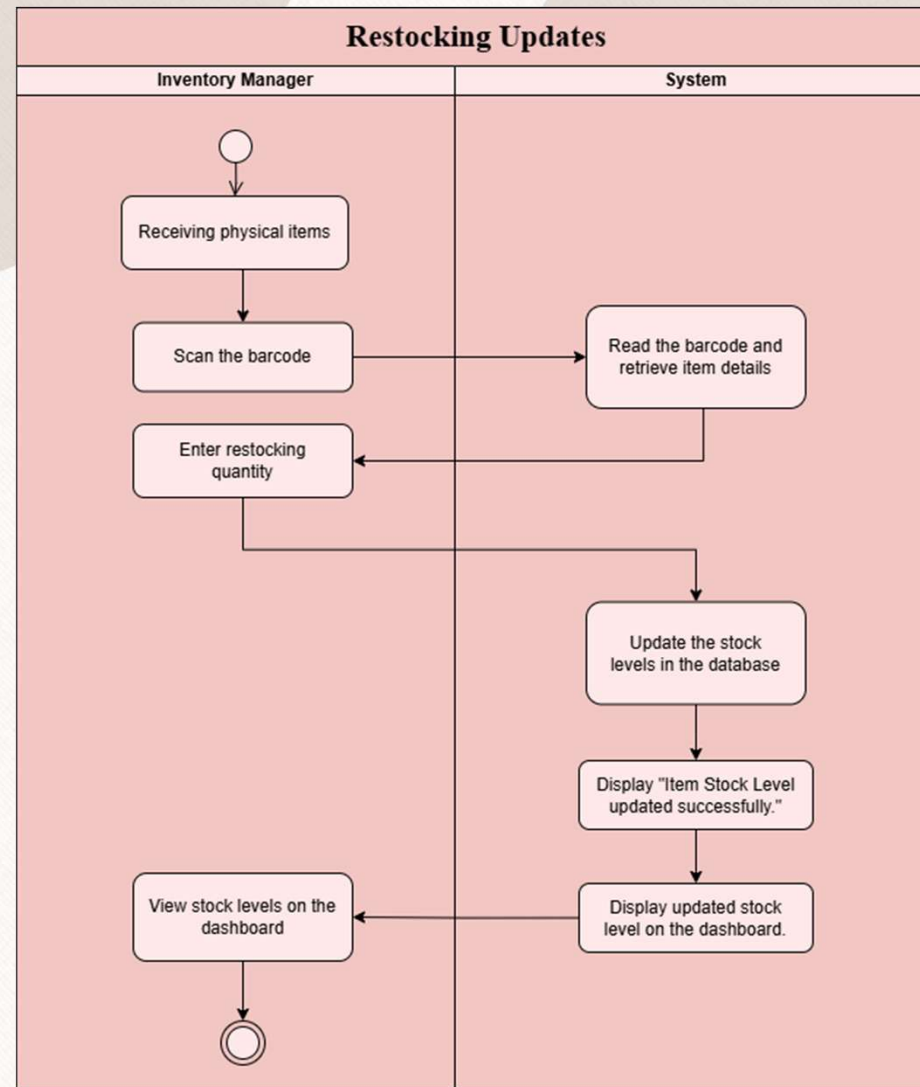
~ Activity Diagram – Withdrawal Updates ~



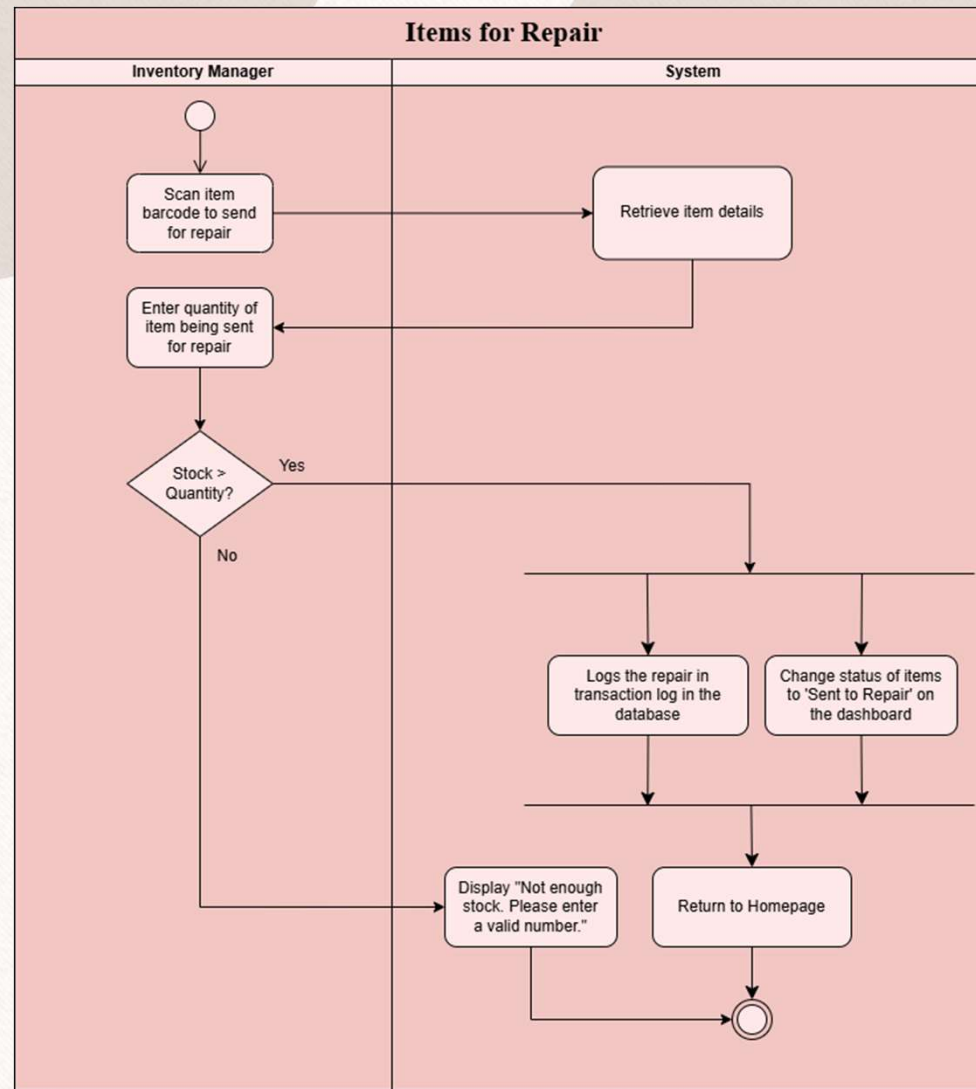
~ Activity Diagram – Ordering Stocks ~



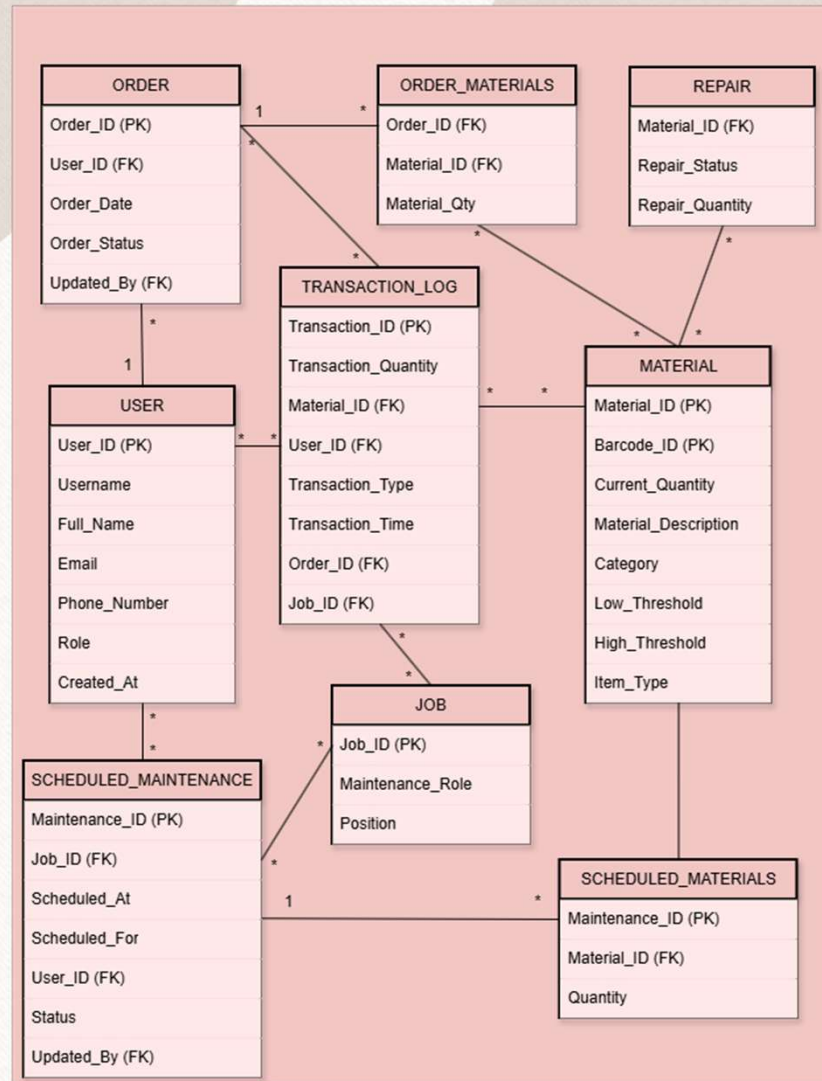
~ Activity Diagram – Restocking Updates ~



~ Activity Diagram – Items Repair Workflow ~



~ Entity Relationship Diagram ~



Role includes Storekeeper,
Maintenance Engineer
and Inventory Manager.

Maintenance_Role includes
Maintenance, Electrical or
Instrumentation.
Position includes Technician
Grade 5, Technician Grade 6,
Technician Grade 7 and
Maintenance Supervisor.

Category includes
Mechanical, Electrical
and Instrumentation

~ Database Design~

TRANSACTION_LOG	
Transaction_ID (PK) :	String
Transaction_Quantity :	Int
Material_ID (FK) :	String
User_ID (FK) :	String
Transaction_Type :	String
Transaction_Time :	Date/Time
Order_ID (FK) :	String
Job_ID (FK) :	String

SCHEDULED_MAINTENANCE	
Maintenance_ID (PK) :	String
Job_ID (FK) :	String
Scheduled_At :	Date/Time
Scheduled_For :	Date/Time
User_ID (FK) :	String
Status :	String
Updated_By (FK) :	String

USER	
User_ID (PK) :	String
Username :	String
Full_Name :	String
Email :	String
Phone_Number :	String
Role :	String
Created_At :	Date/Time

JOB	
Job_ID (PK) :	String
Maintenance_Role :	String
Position :	String

MATERIAL	
Material_ID (PK) :	String
Barcode_ID (PK) :	String
Current_Quantity :	Int
Material_Description :	String
Category :	String
Low_Threshold :	Int
High_Threshold :	Int
Item_Type :	String

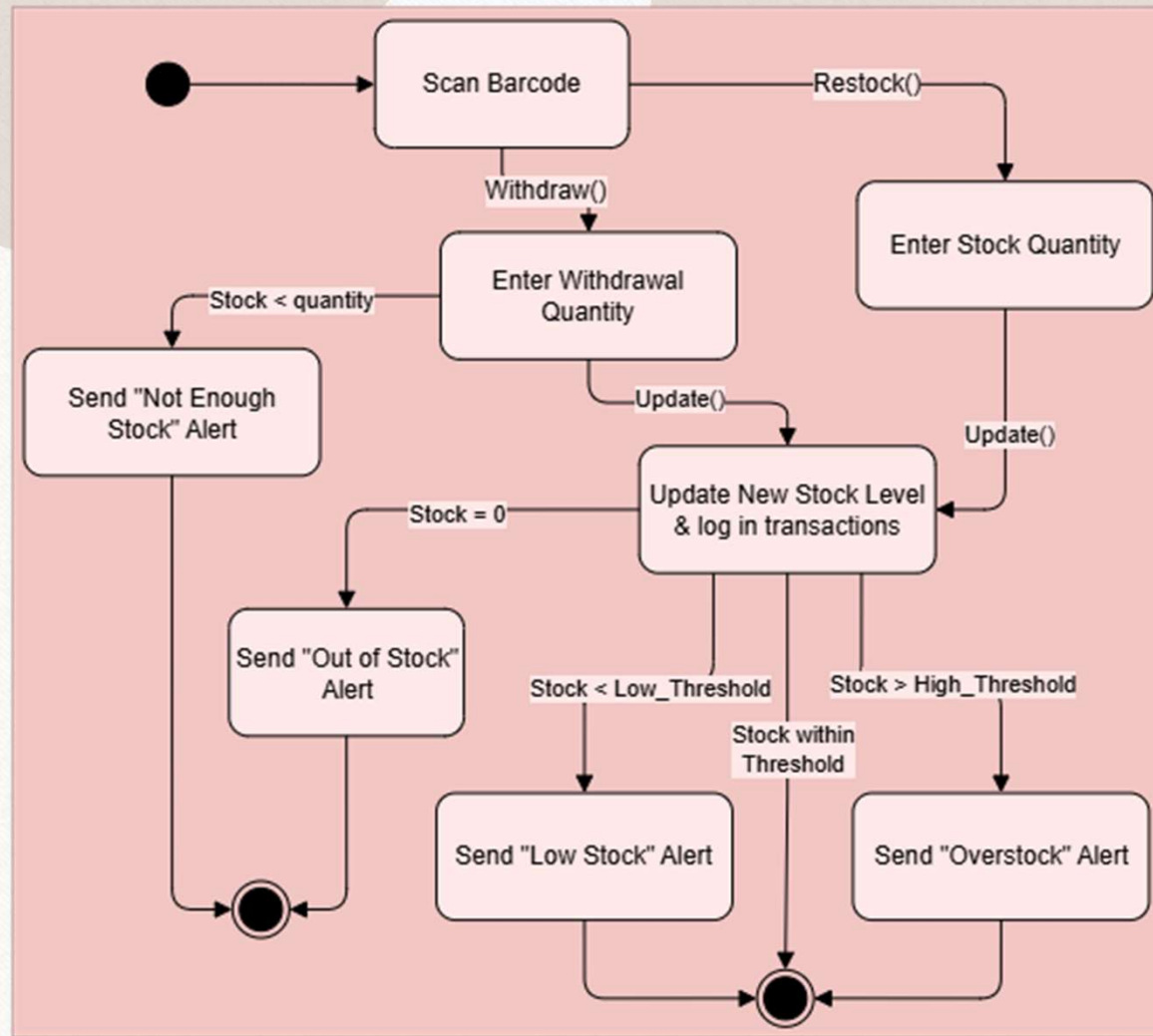
SCHEDULED_MATERIALS	
Maintenance_ID (FK) :	String
Material_ID (FK) :	String
Quantity :	Int

ORDER	
Order_ID (PK) :	String
User_ID (FK) :	String
Order_Date :	Date
Order_Status :	String
Updated_By (FK) :	String

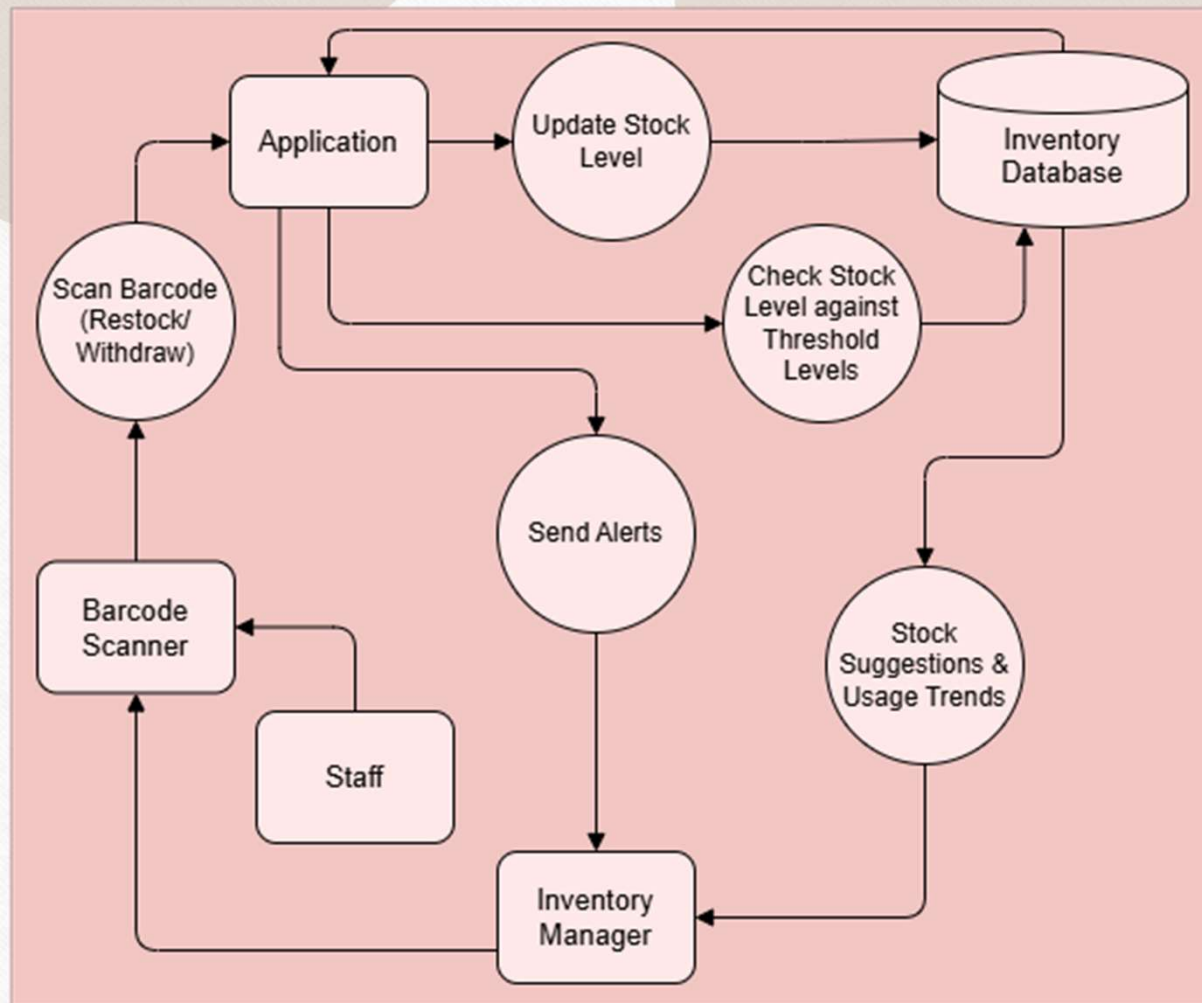
REPAIR	
Material_ID (FK) :	String
Repair_Status :	String
Repair_Quantity :	Int

ORDER_MATERIALS	
Order_ID (FK) :	String
Material_ID (FK) :	String
Material_Qty :	Int

~ Design Diagram – State Diagram ~



~ Design Diagram – Data Flow Diagram ~





Thank you